Assessment of NBTI in Presence of Self-Heating in High-k SOI FinFETs

Udit Monga, Sourabh Khandelwal, Jasmin Aghassi, Josef Sedlmeir, and Tor A. Fjeldly

***Abstract*—We have experimentally investigated the threshold voltage shift due to negative bias temperature instability (NBTI). The NBTI stress in the absence of self-heating (SH) is performed at two different temperatures, i.e., $T = 25$ °C and 125 °C, at bias conditions: gate voltage $V_{gs} = -2$ V and drain voltage $V_{ds} = 0$ V. To evaluate the effect of NBTI in the presence of SH, the stress is performed at room temperature and at $V_{gs} = -2$ V and $V_{ds} = -1$ V. It has been observed that NBTI in the presence of SH causes a significant shift in the threshold voltage.**

***Index Terms*—FinFET, negative bias temperature instability (NBTI), reliability, self-heating (SH), silicon-on-insulator (SOI), thermal resistance.**

I. Introduction

NEGATIVE bias temperature instability (NBTI) is a major concern in nanoscale devices with highly scaled effective oxide thickness (EOT). The electrical stress on the transistor generates traps at the silicon–insulator (SiO_2/SiON) interface and possibly in the insulator bulk. These defects increase the threshold voltage, reduce the channel mobility of the MOSFETs, increase the subthreshold slope, and thus degrade the performance. A significant reduction in the performance metrics implies a shorter device lifetime. Some studies to explain (and model) NBTI have been done [1], [2]; it has been proposed that NBTI in devices with SiON gate stacks is due to breakage of Si–H bonds at elevated temperature and gate voltages [2]. Even in high-k devices, the instigation of NBTI is still considered as breakage of the Si–H bonds due to the existence of an interfacial layer of SiO_2 (approximately 1 nm).

In [3], we presented the effects of self-heating (SH) in silicon-on-insulator (SOI) FinFETs on analog circuits. We showed that SH can have a profound effect on the intrinsic gain. Some other studies on SH in SOI devices have also shown significant impact on performance [4], [5] and reliability [6].

Manuscript received July 11, 2012; revised August 7, 2012; accepted August 9, 2012. Date of publication September 27, 2012; date of current version October 19, 2012. This work was supported by the European Commission under Grant Agreement 218255 (COMON). The review of this letter was arranged by Editor E. A. Gutiérrez-D.

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Digital Object Identifier 10.1109/LED.2012.2213572

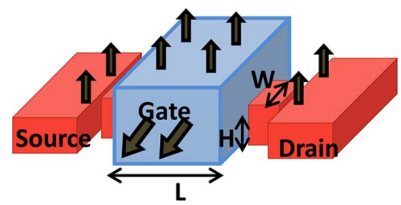

Fig. 1. Heat dissipation paths in the trigate FinFET.

These studies have motivated us to study the effects of SH on NBTI, as NBTI is a temperature-aggravated phenomenon.

Usually, NBTI stress is given for zero drain bias to avoid any hot-carrier injection (HCI), but HCI is not a major concern in PFETs. This is due to the fact that the holes have lower mobility, and thus, they are less effective in impact ionization. In addition, the tunneling probability of the holes is less than that of the electrons. Thus, we believe that the threshold voltage shifts in our measurements as presented for PFETs in this letter mainly from NBTI

II. Temperature Estimation in the Device

An estimation of the average temperature T_d in the device is made using thermal resistance $R_{\rm th}$ and power dissipation P_d

$$T_d = R_{\rm th} P_d + T_a \tag{1}$$

where T_a is the ambient temperature. Power dissipation can be readily obtained from experimental I–V data. The thermal resistance can be extracted from the difference of isothermal and nonisothermal conductance values, and it is given as [7]

$$R_{\rm th} = {\rm Re}\left\{\frac{\Delta Y_{\rm DG}}{(d\,I_D/d\,T_a)V_{ds}Y_{\rm dgiso}}\right\} \tag{2}$$

where $\Delta Y_{\rm DG}$ is the difference between isothermal and nonisothermal transconductance values, and $Y_{\rm dgiso}$ is the isothermal transconductance. The isothermal transconductance and nonisothermal transconductance are obtained from the S-parameters measurements presented in [3]. The extracted thermal resistance is used to make a scalable model by identifying the heat dissipation paths in the device [8], as shown in Fig. 1.

An electrical equivalent of different heat dissipation paths is shown in Fig. 2, where heat generated is represented by the current source and the three conductance values represent the paths of heat dissipation. The amount of heat dissipated from a path is proportional to its cross-sectional area. The equivalent

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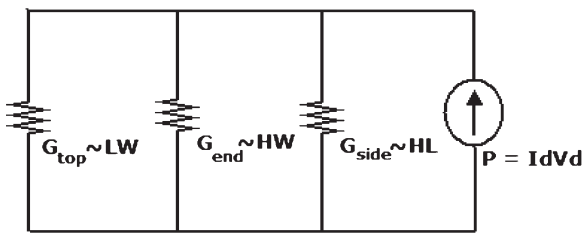

Fig. 2. Electrical equivalent of the heat dissipation paths.

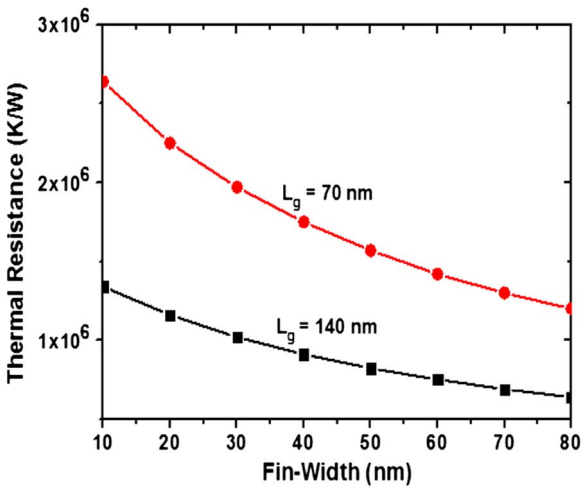

Fig. 3. Variation of thermal resistance of PFETs with the fin width for different channel lengths.

resistance of this network is the thermal resistance of the device (extracted from the measured data), which is given as

$$R_{\text{th}} = \frac{1}{G_{\text{th}}} = \frac{1}{\sigma_{\text{top}}LW + \sigma_{\text{end}}HW + \sigma_{\text{side}}HL} \tag{3}$$

where G_{th} is the net thermal conductance, σ is the heat conductance per unit area for a particular heat dissipation path, L is the gate length, H is the fin height, and W is the fin width (silicon thickness). The heat conductance per unit area, σ_{top}, σ_{end}, and σ_{side} are obtained from the above equation by using the extracted thermal resistance for various device geometries. The above model can be used to predict the net thermal resistance R_{th}, and hence, the temperature rise of any device geometry, but for similar heat conductance values. As apparent from the above equation, R_{th} depends, among other factors, on the active area of a FinFET. Increasing the active area (increasing the gate length or fin width) results in lower thermal resistance. This is evident from the variation of extracted R_{th} (see Fig. 3) from our measurement on PFET devices. The extraction was done at $V_{gs} = V_{ds} = 1.2$ V.

The thermal resistance values are extracted from the data of a FinFET with a SiON gate stack, and our NBTI measurements are performed on devices with a HfO_2 gate stack, but both with the same EOT. We believe that the above model is capable of closely predicting the temperature rise in Hf-based gate stacks because: 1) there is an interfacial layer of SiO_2 of thickness approximately 1 nm, which decreases the overall thermal conductivity; and 2) although HfO_2 has higher thermal conductivity than SiO_2, the physical thickness of HfO_2 is comparatively higher, resulting in an overall thermal conductivity per unit area close to the SiON gate stack.

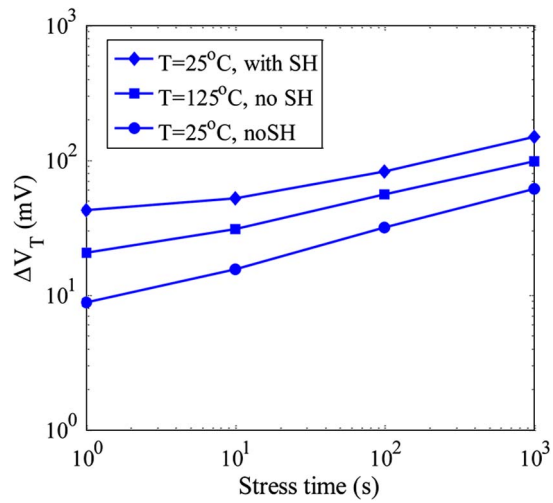

Fig. 4. Variation of V_T shift with stress time for different stressing conditions.

III. Characterization Methodology, Results, and Discussion

A conventional stress–measure–stress technique was used to characterize the NBTI, and a linear threshold voltage V_T shift was measured after each stress phase. Measurements were done on high-k SOI trigate FinFETs with gate length $L = 70$ nm and fin width $W = 10$ nm. The devices were processed with a HfO_2 gate stack and a metal midgap TiN gate; the EOT is 1.9 nm. For NBTI characterization without SH, the devices were stressed with $V_{gs} = -2$ V and zero source/drain bias. The devices were stressed at ambient temperatures $T = 25$ °C and 125 °C. The stress measurements with SH were performed at $V_{gs} = -2$ V, $V_{ds} = -1$ V, and $T = 25$ °C since current flow raises the temperature of the channel. The measurements were performed after four stress times, namely, 1, 10, 100, and 1000 s.

Fig. 4 shows the variation of the V_T shift with stress time for different stressing conditions. V_T was extracted using the constant-current method at $V_{ds} = 50$ mV. As apparent from the figure, the V_T shift is highest for the device stressed with SH. To understand this, it is important to extract the temperature rise during SH stressing. Our measurements indicate that the average drain current at the SH-stress condition ($V_{gs} = -2$ V and $V_{ds} = -1$ V) is 1200 μA/μm. For a single device, this gives the average power dissipation, $P_d = 156$ μW, and approximately a temperature rise to $T \approx 400$ °C, which is significantly higher than for the conventional NBTI stress temperature (125 °C). Hence, the maximum V_T shift is observed in the presence of the SH.

IV. Conclusion

We have experimentally demonstrated a significant impact of SH on NBTI in FinFETs. The conventional stress–measure technique is used to characterize NBTI. It has been assumed that HCI is negligible compared with SH, and thus, the threshold voltage shift occurs mostly from NBTI. The device

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temperature is extracted from the experimental data, and a model for thermal resistance is presented. Our model predicts that the device temperature in the presence of SH is around 400 °C, and thus, it significantly aggravates the NBTI.

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